

Machine Learning for Predicting Primary Graft Dysfunction After Lung Transplantation: An Interpretable Model Study

Wei Xia, MD,¹ Weici Liu, MD,² Zhao He, MD,² Chenghu Song, MD,² Jiwei Liu, MD,² Ruo Chen, MD,² Jingyu Chen, MD,³ Xiaokun Wang, MD,² Hongyang Xu, MD,¹ and Wenjun Mao, MD²

Background. Primary graft dysfunction (PGD) develops within 72 h after lung transplantation (Lung Tx) and greatly influences patients' prognosis. This study aimed to establish an accurate machine learning (ML) model for predicting grade 3 PGD (PGD3) after Lung Tx. **Methods.** This retrospective study incorporated 802 patients receiving Lung Tx between July 2018 and October 2023 (640 in the derivation cohort and 162 in the external validation cohort), and 640 patients were randomly assigned to training and internal validation cohorts in a 7:3 ratio. Independent risk factors for PGD3 were determined by integrating the univariate logistic regression and least absolute shrinkage and selection operator regression analyses. Subsequently, 9 ML models were used to construct prediction models for PGD3 based on selected variables. Their prediction performances were further evaluated. Besides, model stratification performance was assessed with 3 posttransplant metrics. Finally, the SHapley Additive exPlanations algorithm was used to understand the predictive importance of selected variables. **Results.** We identified 9 independent clinical risk factors as selected variables. Among 9 ML models, the random forest (RF) model displayed optimal performance (area under the curve [AUC] = 0.9415, sensitivity [Se] = 0.8972, specificity [Sp] = 0.8795 in the training cohort; AUC = 0.7975, Se = 0.7520, Sp = 0.7313 in the internal validation cohort; and AUC = 0.8214, Se = 0.8235, Sp = 0.6667 in the external validation cohort). Further assessments on calibration and clinical usefulness indicated the promising applicability of the RF model in PGD3 prediction. Meanwhile, the RF model also performed best in terms of risk stratification for postoperative support (extracorporeal membrane oxygenation) (webviewer): $P < 0.001$, mechanical ventilation time: $P = 0.006$, intensive care unit time: $P < 0.001$. **Conclusions.** The RF model had the optimal performance in PGD3 prediction and postoperative risk stratification for patients after Lung Tx.

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¹ Department of Intensive Care Unit, The Affiliated Wuxi People's Hospital of Nanjing Medical University, Wuxi People's Hospital, Wuxi Medical Center, Nanjing Medical University, Wuxi, Jiangsu, China.

² Department of Thoracic Surgery, The Affiliated Wuxi People's Hospital of Nanjing Medical University, Wuxi People's Hospital, Wuxi Medical Center, Nanjing Medical University, Wuxi, Jiangsu, China.

³ Department of Lung Transplantation, The Affiliated Wuxi People's Hospital of Nanjing Medical University, Wuxi People's Hospital, Wuxi Medical Center, Nanjing Medical University, Wuxi, Jiangsu, China.

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W.M., H.X., and J.C. conceived this study. W.X., W.L., and X.W. drafted the article. W.L., Z.H., C.S., J.L., and R.C. collected the data. W.X., W.L., and X.W. performed data analysis. H.X. and W.M. supervised this project. W.X., H.X., and W.M. provided funding support. All authors read and approved the final article.

W.X. and W.L. contributed equally to this work and share first authorship.

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee of the Affiliated Wuxi People's

Hospital of Nanjing Medical University (approval No. KYLLKS201807). The ethics committees waived the need for patient informed consent, in compliance with local legislation on retrospective analyses of de-identified health data.

The data used and/or analyzed during this study cannot be made publicly available due to data privacy regulations but are available from the corresponding author on reasonable request.

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Correspondence: Wenjun Mao, MD, Department of Thoracic Surgery, The Affiliated Wuxi People's Hospital of Nanjing Medical University, Wuxi Medical Center, Nanjing Medical University, No. 299 Qingyang Rd, Wuxi 214023, China. (maowenjun1@njmu.edu.cn); Hongyang Xu, MD, Department of Intensive Care Unit, The Affiliated Wuxi People's Hospital of Nanjing Medical University, Wuxi Medical Center, Nanjing Medical University, No. 299 Qingyang Rd, Wuxi 214023, China. (xhy1912@aliyun.com); Xiaokun Wang, MD, Department of Thoracic Surgery, The Affiliated Wuxi People's Hospital of Nanjing Medical University, Wuxi Medical Center, Nanjing Medical University, No. 299 Qingyang Rd, Wuxi 214023, China. (wangxiaokun@stu.njmu.edu.cn).

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INTRODUCTION

Primary graft dysfunction (PGD) is the product of moderate to severe acute lung injury as the leading complication, which occurs within 72 h after lung transplantation (Lung Tx) and causes increased perioperative mortality in the first 30 d after Lung Tx.^{1,2} The diagnosis and grading of PGD are based on the presence of pulmonary edema on chest X-ray (CRX) and impaired gas exchange function (PaO_2 [PaO_2]/fraction of inspired oxygen [FiO_2] ratio <300).³ Notably, grade 3 PGD (PGD3), the most severe manifestation of PGD, is characterized by a $\text{PaO}_2/\text{FiO}_2$ ratio <200 , CRX infiltrates, or the use of extracorporeal membrane oxygenation (ECMO) at 72 h per International Society for Heart and Lung Transplantation criteria, and its incidence within the first 72 h after Lung Tx is approximately 30%.^{3,4} PGD exerts adverse effects on both short- and long-term outcomes after Lung Tx, including length of hospital stay, ventilator support duration, survival, etc.^{5,6} Hence, perioperative timely identification and effective strategies for preventing and managing PGD are urgently demanded to improve outcomes.

Previous studies have indicated some potential clinical risk factors associated with increased risk of PGD, including donor and recipient characteristics, comorbidities, complications, surgical factors, hematologic indicators, etc.⁷⁻¹⁰ Nonetheless, these factors fail to accurately predict the occurrence of PGD and some may be challenging to effectively use in clinical scenarios. Worse still, it remains elusive how Lung Tx clinicians should consider these variables into account when determining an accurate prognosis and performing decision-making. Recent efforts have focused on integrating a series of selected parameters by traditional regression or machine learning (ML) methods to construct effective prediction models for PGD. Although several studies have established preliminary models to predict PGD after Lung Tx, these models displayed undesirable performance in predicting PGD3 incidence after Lung Tx.¹¹⁻¹⁴ Therefore, it is necessary to develop a well-performing prediction model for patients with PGD3 after Lung Tx. In this study, we developed and validated a promising prediction model for PGD3 occurrence and risk stratification by comparing 9 ML algorithms to assist in posttransplant decision-making.

MATERIALS AND METHODS

Study Design and Participants

Patients receiving Lung Tx at the Affiliated Wuxi People's Hospital of Nanjing Medical University between July 2018 and October 2023 were enrolled, and their clinical information and radiological images were accordingly collected. Patients were included (1) who underwent primary single or bilateral Lung Tx, (2) who were Chinese, (3) who were aged 18 y or older, and (4) whose comprehensive information on perioperative evaluation was available. Patients were excluded (1) who had undergone other kinds of transplantation surgeries; (2) who simultaneously received Lung Tx and surgery on other organs; and (3) who were with severe missing data, defined as missing (a) any one of the key variables such as ECMO, cold ischemia time (CIT), operative time (OT), principal diagnosis, and blood loss and transfusion or (b) $>20\%$ of the other noncritical variables. The study protocol was approved by the Ethics Committee of

the Affiliated Wuxi People's Hospital of Nanjing Medical University (approval No. KYLLKS201807), and informed consent was waived, considering the retrospective nature of this study. This research complied with the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis reporting guideline.¹⁵

Data Collection

For variable selection, patients' clinical information was acquired from medical records as shown in Table 1. The principal diagnosis refers to the primary condition that leads to the eventual Lung Tx. Chronic steroid use means a history of steroid medication use within 3 mo before Lung Tx. The ECMO type consists of venovenous cannulation ECMO and venoarterial cannulation ECMO. Additionally, for missing data that do not involve key variables, we used K-nearest neighbor imputation to fill in the gaps.

For outcome definitions and measurements, the primary outcome was the development of PGD3 within 72 h post-Lung Tx. PGD was defined and graded according to the latest International Society for Heart and Lung Transplantation consensus criteria.³ PGD3 is characterized by $\text{PaO}_2/\text{FiO}_2 <200$, CRX infiltrates, and/or the use of ECMO at 72 h.³ The PaO_2 and FiO_2 of patients were recorded at 4 specific time points: T0 (within 6 h of past lung reperfusion, the data within 2 h of arrival at the intensive care unit [ICU] used in this study was compliant), T24, T48, and T72, respectively. Patients who met the criteria at any of these time points were diagnosed with PGD3. The secondary outcomes included ICU stay duration, ECMO support duration, and mechanical ventilation (MV) duration after Lung Tx. The ICU stay duration was defined as the time from admission to the ICU after transplantation until discharge to a general ward. The ECMO support duration was calculated from the start of ECMO to successful weaning or discontinuation. The MV duration was measured from intubation in the operating room to successful extubation in the ICU. For patients requiring reintubation, the cumulative time on MV was recorded. These secondary outcomes were selected to evaluate post-transplant clinical course and potential complications.

Variable Selection

To identify the variables that can be included in the ML models, the univariate logistic regression and least absolute shrinkage and selection operator (LASSO) regression analyses were used to determine independent risk factors for PGD3. Their integration could minimize redundancy and enhance model parsimony. Variables with statistical significance ($P < 0.05$) in logistic regression analysis were incorporated, and odds ratios and 95% confidence intervals were estimated. LASSO regression analysis, using the L1 regularization approach, established inclusion and exclusion criteria according to the magnitude of coefficient variables. The addition of an L1 penalty term in LASSO regression to the loss function drives certain coefficients to zero, thereby facilitating automatic feature selection. Independent variables with zero coefficients were excluded, whereas those with nonzero coefficients were retained in LASSO regression. The intersection of logistic regression and LASSO results yielded a refined set of variables for subsequent ML input.

TABLE 1.**Clinical characteristics of non-PGD3 and PGD3 patients after lung transplantation**

Characteristics	Derivation cohort				External validation cohort			
	Non-PGD3 (N = 407)	PGD3 (N = 233)	Total (N = 640)	P	Non-PGD3 (N = 102)	PGD3 (N = 60)	Total (N = 162)	P
Age, y, mean ± SD	53.49 ± 13.52	53.59 ± 14.09	53.53 ± 13.72	0.788	54.50 ± 12.05	52.17 ± 13.95	53.64 ± 12.79	0.371
Height, cm, mean ± SD	166.51 ± 6.43	166.39 ± 6.35	166.46 ± 6.40	0.684	167.63 ± 6.54	167.32 ± 5.87	167.51 ± 6.29	0.743
Weight, kg, mean ± SD	58.06 ± 11.36	57.15 ± 11.22	57.73 ± 11.31	0.489	55.50 ± 10.29	63.08 ± 10.17	58.31 ± 10.85	<0.001
BMI, kg/m ² , mean ± SD	21.01 ± 4.34	20.77 ± 4.54	20.92 ± 4.41	0.631	19.82 ± 3.83	22.59 ± 3.82	20.85 ± 4.04	<0.001
Sex, n (%)				0.913				0.242
Male	308 (75.7)	178 (76.4)	486 (75.9)		87 (85.3)	46 (76.7)	133 (82.1)	
Female	99 (24.3)	55 (23.6)	154 (24.1)		15 (14.7)	14 (23.3)	29 (17.9)	
Principal diagnosis, n (%)				0.055				0.002
COPD	73 (17.9)	27 (11.6)	100 (15.6)		23 (22.5)	2 (3.3)	25 (15.4)	
Bronchiectasis	16 (3.9)	10 (4.3)	26 (4.1)		4 (3.9)	0 (0.0)	4 (2.5)	
Obstructive bronchiolitis	7 (1.7)	1 (0.4)	8 (1.3)		1 (1.0)	2 (3.3)	3 (1.9)	
Fibrosis	219 (53.8)	132 (56.7)	351 (54.8)		42 (41.2)	41 (68.3)	83 (51.2)	
Silicosis	61 (15.0)	35 (15.0)	96 (15.0)		20 (19.6)	8 (13.3)	28 (17.3)	
PAH	5 (1.2)	6 (2.6)	11 (1.7)		2 (2.0)	4 (6.7)	6 (3.7)	
GVHD	9 (2.2)	14 (6.0)	23 (3.6)		8 (7.8)	2 (3.3)	10 (6.1)	
Others	17 (4.2)	8 (3.4)	25 (3.9)		2 (2.0)	1 (1.7)	3 (1.9)	
Smoking history, n (%)				0.823				0.192
No	236 (58.0)	138 (59.2)	374 (58.4)		56 (54.9)	40 (66.7)	96 (59.3)	
Yes	171 (42.0)	95 (40.8)	266 (41.6)		46 (45.1)	20 (33.3)	66 (40.7)	
Alcohol consumption, n (%)				0.354				0.512
No	386 (94.8)	216 (92.7)	602 (94.1)		96 (94.1)	54 (90.0)	150 (92.6)	
Yes	21 (5.2)	17 (7.3)	38 (5.9)		6 (5.9)	6 (10.0)	12 (7.4)	
Diabetes, n (%)				0.408				0.097
No	336 (82.6)	199 (85.4)	535 (83.6)		87 (85.3)	44 (73.3)	131 (80.9)	
Yes	71 (17.4)	34 (14.6)	105 (16.4)		15 (14.7)	16 (26.7)	31 (19.1)	
CHD, n (%)				0.518				0.461
No	353 (86.7)	197 (84.5)	550 (85.9)		89 (86.7)	41 (68.3)	130 (81.7)	
Yes	54 (13.3)	36 (15.5)	90 (14.1)		13 (12.7)	19 (31.7)	32 (19.8)	
HBP, n (%)				0.143				0.627
No	332 (81.6)	178 (76.4)	510 (79.7)		86 (84.3)	48 (80.0)	134 (82.7)	
Yes	75 (18.4)	55 (23.6)	130 (20.3)		16 (15.7)	12 (20.0)	28 (17.3)	
PAH, n (%)				0.317				0.640
No	198 (48.6)	103 (44.2)	301 (47.0)		46 (45.1)	24 (40.0)	70 (43.2)	
Yes	209 (51.4)	130 (55.8)	339 (53.0)		56 (54.9)	36 (60.0)	92 (56.8)	
Cardiac insufficiency, n (%)				0.966				0.521
No	346 (85.0)	197 (84.5)	543 (84.8)		90 (88.2)	50 (83.3)	140 (86.4)	
Yes	61 (15.0)	36 (15.5)	97 (15.2)		12 (11.8)	10 (16.7)	22 (13.6)	
6MWT, m, n (%)				0.055				0.190
<150	171 (42.0)	114 (48.9)	285 (44.5)		57 (55.9)	39 (65.0)	96 (59.3)	
150–425	226 (55.5)	118 (50.6)	344 (53.8)		45 (44.1)	20 (33.3)	65 (40.1)	
>425	10 (2.5)	1 (0.4)	11 (1.7)		0 (0.0)	1 (1.7)	1 (0.6)	
Hb, g/L, mean ± SD	129.28 ± 19.59	130.60 ± 20.26	129.76 ± 19.83	0.266	130.84 ± 20.19	130.63 ± 23.48	130.76 ± 21.40	0.718
FVC, mL, mean ± SD	1.35 ± 0.74	1.38 ± 0.79	1.36 ± 0.76	0.837	1.48 ± 0.79	1.92 ± 1.59	1.64 ± 1.17	0.063
FVC%, mean ± SD	0.36 ± 0.20	0.38 ± 0.20	0.37 ± 0.20	0.395	0.39 ± 0.13	0.43 ± 0.17	0.41 ± 0.15	0.116
FEV ₁ , g/L, mean ± SD	0.95 ± 0.62	0.95 ± 0.66	0.95 ± 0.63	0.668	1.08 ± 0.55	1.22 ± 0.67	1.13 ± 0.60	0.157
FEV ₁ %, mean ± SD	0.32 ± 0.21	0.32 ± 0.23	0.32 ± 0.22	0.934	0.34 ± 0.16	0.39 ± 0.18	0.36 ± 0.17	0.047
Prior lung surgery (non- transplant), n (%)				0.563				0.534
No	385 (94.6)	217 (93.1)	602 (94.1)		97 (95.1)	59 (98.3)	156 (96.3)	
Yes	22 (5.4)	16 (6.9)	38 (5.9)		5 (4.9)	1 (1.7)	6 (3.7)	
Prior lung transplant, n (%)				0.014				0.741

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TABLE 1. (Continued)

Characteristics	Derivation cohort			P	External validation cohort			P
	Non-PGD3 (N = 407)	PGD3 (N = 233)	Total (N = 640)		Non-PGD3 (N = 102)	PGD3 (N = 60)	Total (N = 162)	
No	401 (98.5)	221 (94.8)	622 (97.2)		98 (96.1)	59 (98.3)	157 (96.9)	
Yes	6 (1.5)	12 (5.2)	18 (2.8)		4 (3.9)	1 (1.7)	5 (3.1)	
Prior TB, n (%)				0.995				0.654
No	370 (90.9)	211 (90.6)	581 (90.8)		88 (86.3)	54 (90.0)	142 (87.7)	
Yes	37 (9.1)	22 (9.4)	59 (9.2)		14 (13.7)	6 (10.0)	20 (12.3)	
Chronic steroid use, n (%)				0.021				0.714
No	225 (55.3)	106 (45.5)	331 (51.7)		40 (39.2)	21 (35.0)	61 (37.7)	
Yes	182 (44.7)	127 (54.5)	309 (48.3)		62 (60.8)	39 (65.0)	101 (62.3)	
Preoperative infection, n (%)				0.999				0.579
No	154 (37.8)	88 (37.8)	242 (37.8)		45 (44.1)	23 (38.3)	68 (42.0)	
Yes	253 (62.2)	145 (62.2)	398 (62.2)		57 (55.9)	37 (61.7)	94 (58.0)	
Type of transplant, n (%)				0.860				0.410
Left	79 (19.4)	49 (21.0)	128 (20.0)		19 (18.6)	8 (13.3)	27 (16.7)	
Right	87 (21.4)	47 (20.2)	134 (20.9)		24 (23.5)	11 (18.3)	35 (21.6)	
Double	241 (59.2)	137 (58.8)	378 (59.1)		59 (57.8)	41 (68.3)	100 (61.7)	
ECMO, n (%)				<0.001				<0.001
No	115 (28.3)	36 (15.5)	151 (23.6)		5 (4.9)	0 (0.0)	5 (3.1)	
Yes	292 (71.7)	197 (84.5)	489 (76.4)		97 (95.1)	60 (100.0)	157 (96.9)	
ECMO type, n (%)				0.001				<0.001
None	115 (28.3)	36 (15.5)	151 (23.6)		5 (4.9)	0 (0.0)	5 (3.1)	
V-A	85 (20.9)	57 (24.5)	142 (22.2)		18 (17.6)	15 (25.0)	33 (20.4)	
V-V	207 (50.9)	140 (60.1)	347 (54.2)		79 (77.5)	45 (75.0)	124 (76.5)	
CIT, h, mean $\pm$ SD	7.22 $\pm$ 1.88	9.08 $\pm$ 2.44	7.89 $\pm$ 2.29	<0.001	7.75 $\pm$ 1.45	8.38 $\pm$ 1.74	7.98 $\pm$ 1.59	0.054
OT, h, mean $\pm$ SD	5.72 $\pm$ 1.72	6.06 $\pm$ 1.84	5.84 $\pm$ 1.77	0.026	2.47 $\pm$ 2.18	3.56 $\pm$ 3.15	2.87 $\pm$ 2.63	0.014
Blood loss volume, mL, median (IQR)	1000 (600–1500)	1000 (800–1500)	1000 (700–1500)	0.093	1000 (800–2125)	950 (600–1500)	1000 (650–1500)	0.007
Transfusion, n (%)				0.007				0.931
None	90 (22.1)	35 (15.0)	125 (19.5)		12 (11.8)	0	18 (11.1)	
Autologous transfusion	8 (2.0)	0 (0.0)	8 (1.3)		0 (0.0)	0 (0.0)	0 (0.0)	
Allogeneic transfusion	309 (75.9)	198 (85.0)	507 (79.2)		90 (88.2)	54 (90.0)	144 (88.9)	
Transfusion volume, mL, median (IQR)	1237.5 (925–1800)	1400 (950–2150)	1350 (925–1993.75)	0.071	1862.5 (1212.5– 3106.25)	1250 (881.25–1625)	1400 (1000–2443.75)	<0.001
Transfusion (RBC), n (%)				0.011				0.719
No	142 (34.9)	58 (24.9)	200 (31.3)		26 (25.5)	13 (21.7)	39 (24.1)	
Yes	265 (65.1)	175 (75.1)	440 (68.7)		76 (74.5)	47 (78.3)	123 (75.9)	
Transfusion volume (RBC), mL, median (IQR)	1050 (800–1600)	1300 (800–2100)	1200 (800–1800)	0.012	1850 (1137.5– 2887.5)	1200 (800–1600)	1325 (900–2375)	<0.001
Transfusion (PRBC), n (%)				0.299				0.754
No	378 (92.9)	222 (95.3)	600 (93.8)		93 (91.2)	53 (88.3)	146 (90.1)	
Yes	29 (7.1)	11 (4.7)	40 (6.2)		9 (8.8)	7 (11.7)	16 (9.9)	
Transfusion volume (PRBC), mL, median (IQR)	800 (600–1000)	600 (400–1000)	800 (575–1050)	0.311	1000 (700–2700)	1400 (800–1600)	1300 (750–2350)	0.558
Transfusion (plasma), n (%)				0.322				0.513
No	295 (72.5)	178 (76.4)	473 (73.9)		83 (81.4)	52 (86.7)	135 (83.3)	
Yes	112 (27.5)	55 (23.6)	167 (26.1)		19 (18.6)	8 (13.3)	27 (16.7)	
Transfusion volume (plasma), mL, median (IQR)	600 (487.5–900)	700 (475–975)	600 (475–925)	0.531	750 (637.5–1575)	600 (550–1112.5)	650 (600–1112.5)	0.173
Transfusion (cryo), n (%)				0.573				0.985
No	399 (98.0)	226 (97.0)	625 (97.7)		100 (98.0)	58 (96.7)	158 (97.5)	
Yes	8 (2.0)	7 (3.0)	15 (2.3)		2 (2.0)	2 (3.3)	4 (2.5)	
Transfusion volume (cryo), mL, median (IQR)	450 (318.75–500)	450 (275–675)	450 (300–500)	0.768	652.5 (551.25– 753.75)	742.5 (596.25–888.75)	652.5 (450–900)	0.999

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TABLE 1. (Continued)

Characteristics	Derivation cohort				External validation cohort			
	Non-PGD3 (N = 407)	PGD3 (N = 233)	Total (N = 640)	P	Non-PGD3 (N = 102)	PGD3 (N = 60)	Total (N = 162)	P
Transfusion (platelet), n (%)				0.999				0.985
No	403 (99.0)	231 (99.1)	634 (99.1)		100 (98.0)	58 (96.7)	158 (97.5)	
Yes	4 (1.0)	2 (0.9)	6 (0.9)		2 (2.0)	2 (3.3)	4 (2.5)	
Transfusion volume (platelet), mL, median (IQR)	250 (250–250)	375 (192 webviewer 50–437.5)	250 (50–437.5)	0.802	625 (437.5–812.5)	375 (312.5–437.5)	375 (250–625)	0.999
Postoperative ECMO support, n (%)				<0.001				0.094
No	132 (32.4)	42 (18.0)	174 (27.2)		7 (6.9)	0 (0)	7 (4.3)	
Yes	275 (67.6)	191 (82.0)	466 (72.8)		95 (93.1)	60 (100.0)	155 (95.7)	
ECMO support duration, d, median (IQR)	1 (0–1)	2 (1–4)	1 (0–2)	<0.001	1 (1–2)	3 (2–5)	1 (1–3)	<0.001
Postoperative ICU stay duration, d, median (IQR)	4 (3–6)	7 (4–13)	5 (3–8)	<0.001	4 (3–6)	8 (5–16)	5 (4–8)	<0.001
Postoperative MV dura- tion, d, median (IQR)	2 (1–3)	4 (2–9)	2 (2–4)	<0.001	2 (2–4)	6 (3–11)	3 (2–6)	<0.001

6MWT, 6-min walk test; BMI, body mass index; CHD, coronary heart disease; CIT, cold ischemia time; COPD, chronic obstructive pulmonary disease; cryo, cryoprecipitate; ECMO, extracorporeal membrane oxygenation; FEV₁, forced expiratory volume at 1 s; FEV₁%, FEV₁ percent predicted; FVC, forced vital capacity; FVC%, FVC percent predicted; GVHD, graft-versus-host disease; Hb, hemoglobin; HBP, high blood pressure; ICU, intensive care unit; IQR, interquartile range; MV, mechanical ventilation; OT, operative time; PAH, pulmonary arterial hypertension; PGD3, grade 3 primary graft dysfunction; PRBC, packed red blood cells; TB, tuberculosis; V-A, venoarterial cannulation; V-V, venovenous cannulation.

Model Development and Validation

The ML models were constructed using R version 4.4.0, in combination with the Tidymodels package designed for ML and adhering to tidy principles. In our study, 9 ML algorithms were leveraged, via the Tidymodels package, to establish PGD3 prediction models. The models were constructed below: decision tree (DT, `decision_tree` function, “`rpart`” engine), elastic net (Enet, `logistic_reg` function, “`glmnet`” engine), *K*-nearest neighbors (KNN, `nearest_neighbor` function, “`knn`” engine), light gradient boosting machine (Lightgbm, `boost_tree` function, “`lightgbm`” engine), logistic regression (logistic, `logistic_reg` function, “`glm`” engine), multilayer perceptron (MLP, `mlp` function, “`mlp`” engine), random forest (RF, `rand_forest` function, “`randomForest`” engine), radial support vector machine (`svm_rbf` function, “`kernlab`” engine), and extreme gradient boosting (Xgboost, `boost_tree` function, “`xgboost`” engine).

Hyperparameter tuning for each ML algorithm was performed using an internal 5-fold cross-validation procedure. After hyperparameter optimization, we retrained the models on the entire training cohort to finalize the weighting and determine a locked model for subsequent assessment on the internal validation cohort. The trained models’ performance was further evaluated by comparing the area under the curve (AUC) of the receiver operator characteristic curve. The calibration curve and decision curve analysis were also performed to evaluate the calibration and net benefits of models. Finally, we assessed the generalization ability of constructed models in the external validation cohort. Based on the cutoff value of the optimal F1 score, model stratification performance was assessed with 3 post-Lung Tx metrics—ICU stay duration, ECMO support duration, and MV duration. The SHAP (SHapley Additive exPlanations) algorithm was used to depict the predictive

importance of selected variables in the optimal PGD3 prediction model and provide global and local explanations.

Statistical Analysis

Baseline data were analyzed to assess the association between patient characteristics and patients with PGD3. Normally distributed continuous variables were reported as mean ± SD and compared between groups using independent samples *t* tests. Nonnormally distributed continuous variables were reported as median (P25 and P75) and compared with Mann-Whitney *U* tests. Categorical variables were reported as frequency (percentage, %), with chi-square tests used for statistical analysis. All statistical tests were 2-sided, and a *P* value of <0.05 was considered statistically significant. Statistical analyses were performed using R version 4.4.0 and Python version 3.11.5.

RESULTS

Characteristics of Patients

As shown in Figure 1, 640 of 710 Lung Tx patients were eventually incorporated in the derivation cohort and randomly assigned to training and internal validation cohorts in a 7:3 ratio for subsequent variable selection and model development. Moreover, 162 of 183 Lung Tx patients were considered eligible and thus included for final external validation that aimed to perform an era-specific comparison from the same hospital.

In the derivation cohort, 233 patients (36.4%) ended up with PGD3 at 72 h after Lung Tx. We summarized the detailed baseline characteristics of patients in the PGD3 and non-PGD3 groups in Table 1. Most patients had a diagnosis of fibrosis (54.8%), followed by chronic obstructive pulmonary disease (15.6%) and silicosis

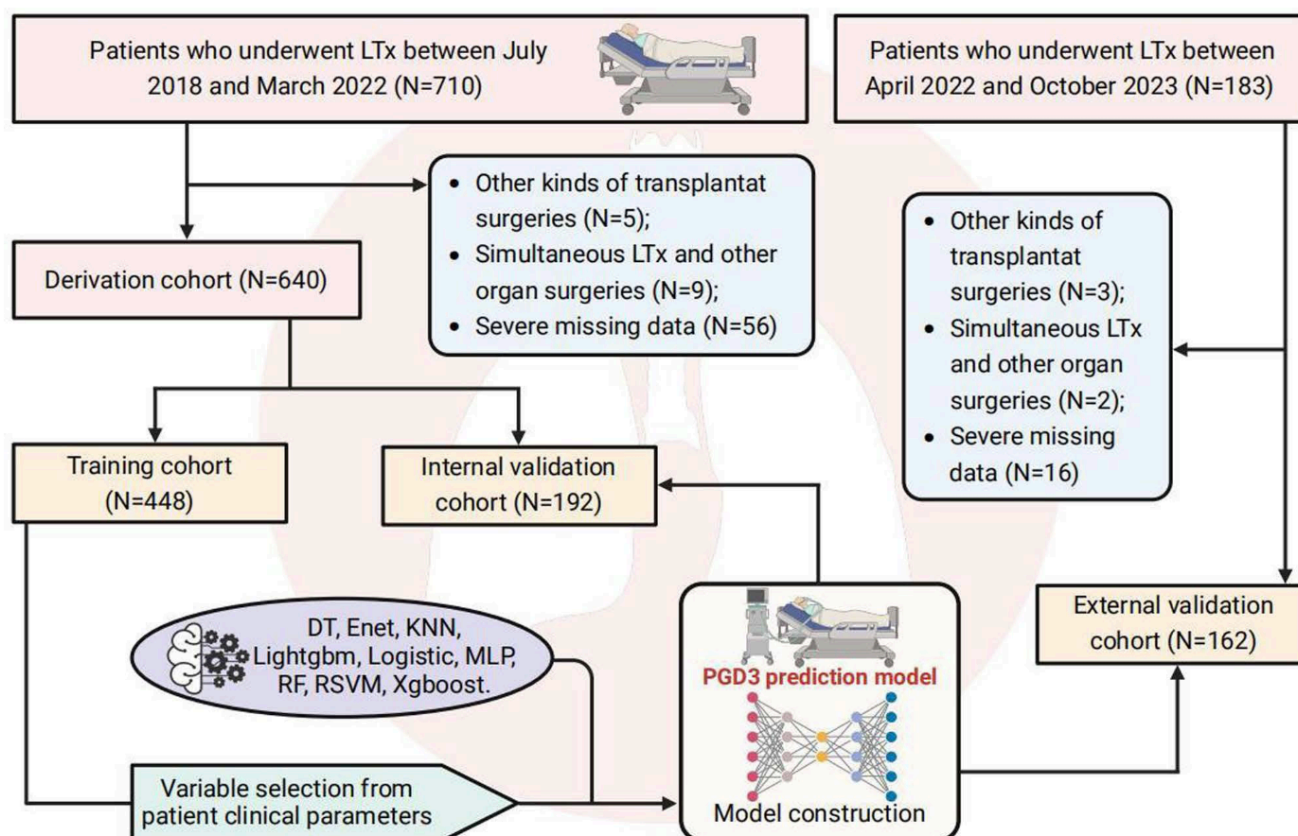


FIGURE 1. The flowchart of Lung Tx patient enrollment. This retrospective study incorporated a total of 802 patients receiving Lung Tx between July 2018 and October 2023 (640 in the derivation cohort and 162 in the external validation cohort), and 640 patients were randomly assigned to training and internal validation cohorts in a 7:3 ratio. The variable selection process was performed only within the training cohort, and the selected variables were used as machine learning inputs for subsequent validation. DT, decision tree; KNN, K-nearest neighbor; Lightgbm, light gradient boosting machine; Logistic, logistic regression; Lung Tx, lung transplantation; MLP, multilayer perceptron; PGD3, grade 3 primary graft dysfunction; RF, random forest; RSVM, radial support vector machine; Xgboost, extreme gradient boosting.

(15.0%). No significant differences were observed in comorbidities among the study population. The 6-min walk test results and multiple pulmonary function parameters indicated impaired cardiopulmonary capacity in most patients, with no significant differences between groups in terms of lung function. Recipients of second Lung Tx exhibited a higher propensity for developing PGD3 compared with first-time recipients ($P = 0.014$), and chronic steroid use was less common in the PGD3 group ($P = 0.021$). Most patients received bilateral Lung Tx, and ECMO was more frequently used in the PGD3 group ($P < 0.001$), especially the venovenous ECMO type ($P = 0.001$). Statistically significant differences in CIT (7.22 ± 1.88 versus 9.08 ± 2.44 , $P < 0.001$) and OT (5.72 ± 1.72 versus 6.06 ± 1.84 , $P = 0.026$) were observed between non-PGD3 and PGD3 groups. We also found significant differences in the transfusion rate for autologous or allogeneic blood between 2 groups ($P = 0.007$) and a higher trend of transfusion volume in the PGD3 group ($P = 0.071$). Notably, the red blood cell (RBC) transfusion rate was higher in the PGD3 group ($P = 0.011$), as was the RBC transfusion volume ($P = 0.012$). Additionally, no significant differences were found in transfusion rates or volumes for packed RBCs, plasma, cryoprecipitate, or platelets between 2 groups. Postoperatively, patients with PGD3 were more likely to require ECMO support ($P < 0.001$) and significantly

prolonged supportive treatments (ECMO support duration, ICU stay duration, and MV duration).

Among these 640 enrolled patients, 448 and 192 patients were randomly assigned to training and internal validation cohorts, respectively. The detailed baseline characteristics of patients in the training and internal validation cohorts are shown in Table S1 (SDC, <http://links.lww.com/TP/D224>), and no significant differences were observed between 2 cohorts. In the training and internal validation cohorts, 166 (37.1%) and 67 (34.9%) patients were categorized as PGD3, respectively, and the difference between 2 cohorts was not statistically significant ($P = 0.667$).

Similarly, in the external validation cohort, 60 patients (37.0%) had PGD3 at 72 h after Lung Tx. Table 1 also displayed the baseline characteristics of patients in the PGD3 and non-PGD3 groups. It was noteworthy that statistically significant differences could be found in pulmonary function test results (eg, forced vital capacity, forced vital capacity percent, forced expiratory volume at 1 s, forced expiratory volume at 1 s percent, 6-min walk test), chronic steroid use, ECMO type, CIT, OT, and so on ($P < 0.05$) between the derivation and external validation cohorts (Table S1, SDC, <http://links.lww.com/TP/D224>), suggesting that the external validation cohort was to some extent heterogeneous from the derivation cohort and eligible to evaluate generalization ability of the model.

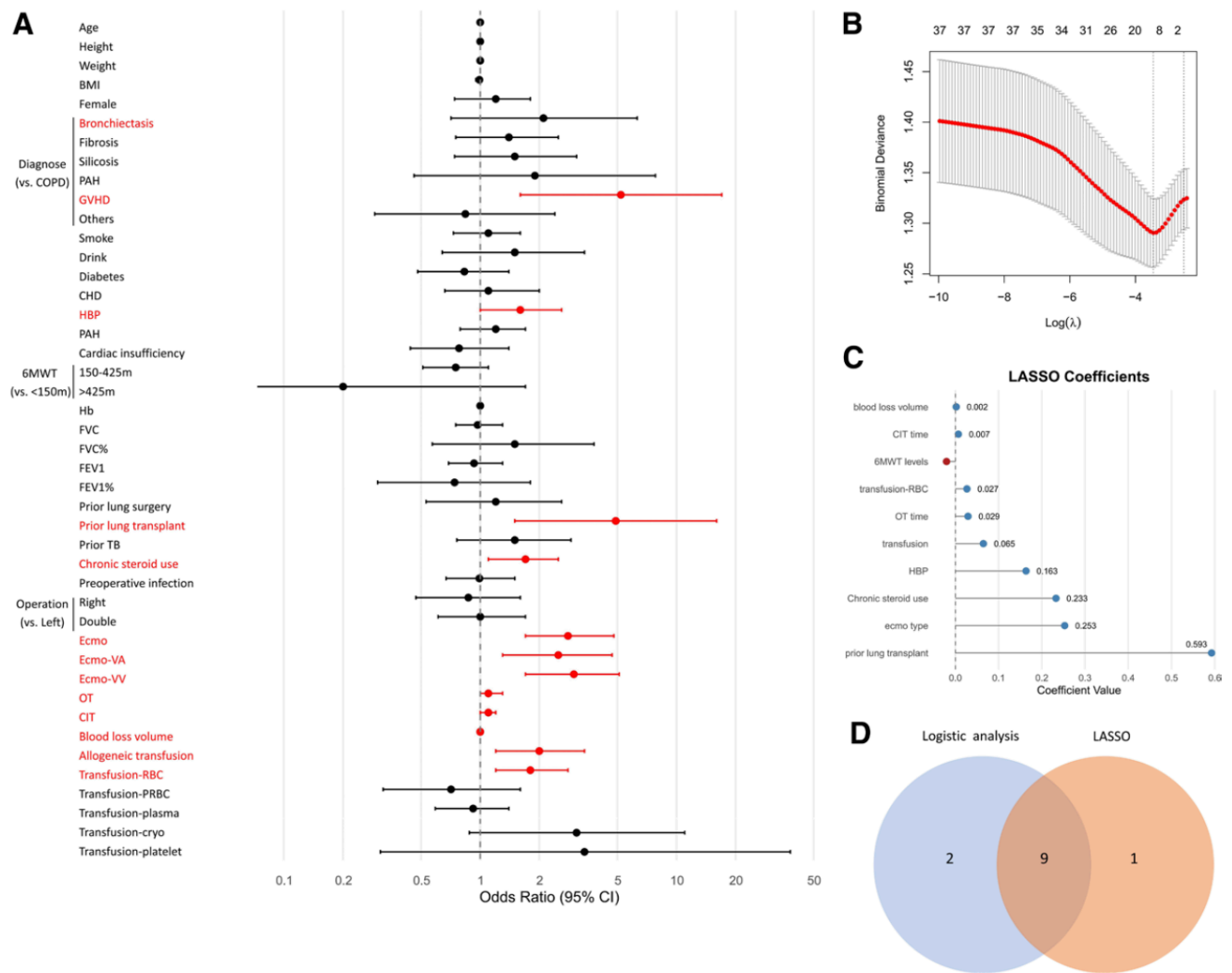


FIGURE 2. Variable selection process of univariate logistic regression analysis and LASSO regression analysis. A, Univariate logistic regression analysis of clinical parameters. B, LASSO regression analysis of clinical parameters. C, Feature visualization of LASSO regression. D, Venn diagram showing the intersection of clinical parameters of univariate logistic regression and LASSO regression. 6MWT, 6-min walk test; BMI, body mass index; CHD, coronary heart disease; CI, confidence interval; CIT, cold ischemia time; COPD, chronic obstructive pulmonary disease; cryo, cryoprecipitate; ECMO, extracorporeal membrane oxygenation; FEV₁, forced expiratory volume at 1 s; FEV₁%, FEV₁ percent predicted; FVC, forced vital capacity; FVC%, FVC percent predicted; GVHD, graft-vs-host disease; Hb, hemoglobin; HBP, high blood pressure; LASSO, least absolute shrinkage and selection operator; OT, operative time; PAH, pulmonary arterial hypertension; PRBC, packed red blood cell; TB, tuberculosis; V-A, venoarterial cannulation; V-V, venovenous cannulation.

Variable Selection

To optimize model efficiency and applicability, we conducted a logistic regression analysis on independent variables (Figure 2A), aiming to distill a more concise and accurate set of variables. Results showed that 11 indicators were independent risk factors for PGD3. LASSO regression optimization used 10-fold cross-validation, yielding an optimal tuning parameter (λ) of 0.032 (Figure 2B). Through LASSO regression variable selection, 10 independent variables were selected from the initial variable pool (Figure 2C; Table S2, SDC, <http://links.lww.com/TP/D224>). LASSO regression and logistic regression yielded concordant feature selection results, reinforcing the robustness of the identified predictors. The overlap between 2 analysis methods revealed 9 variables as input features in ML models, including ECMO type, chronic steroid use, prior Lung Tx, high blood pressure (HBP), allogeneic transfusion, RBC transfusion, OT,

CIT, and blood loss volume (Figure 2D; Figure S1, SDC, <http://links.lww.com/TP/D224>). Additionally, the analysis of multicollinearity for 9 variables displayed the variance inflation factor value <2, signifying the absence of notable multicollinearity concerns (Table S3, SDC, <http://links.lww.com/TP/D224>).

Development and Validation of the PGD3 Prediction Model

In the model training, PGD3 presence was encoded as the positive class, with its absence as the negative class. Using these 9 features, we developed 9 different ML models, including DT, Enet, KNN, Lightgbm, logistic, MLP, RF, radial support vector machine, and Xgboost. Results indicated that the RF model showed a significantly higher AUC value in comparison with the rest 8 ML algorithms across training (Figure 3A–C) and internal validation cohorts (Figure 3D–F). Furthermore, a series of parameters

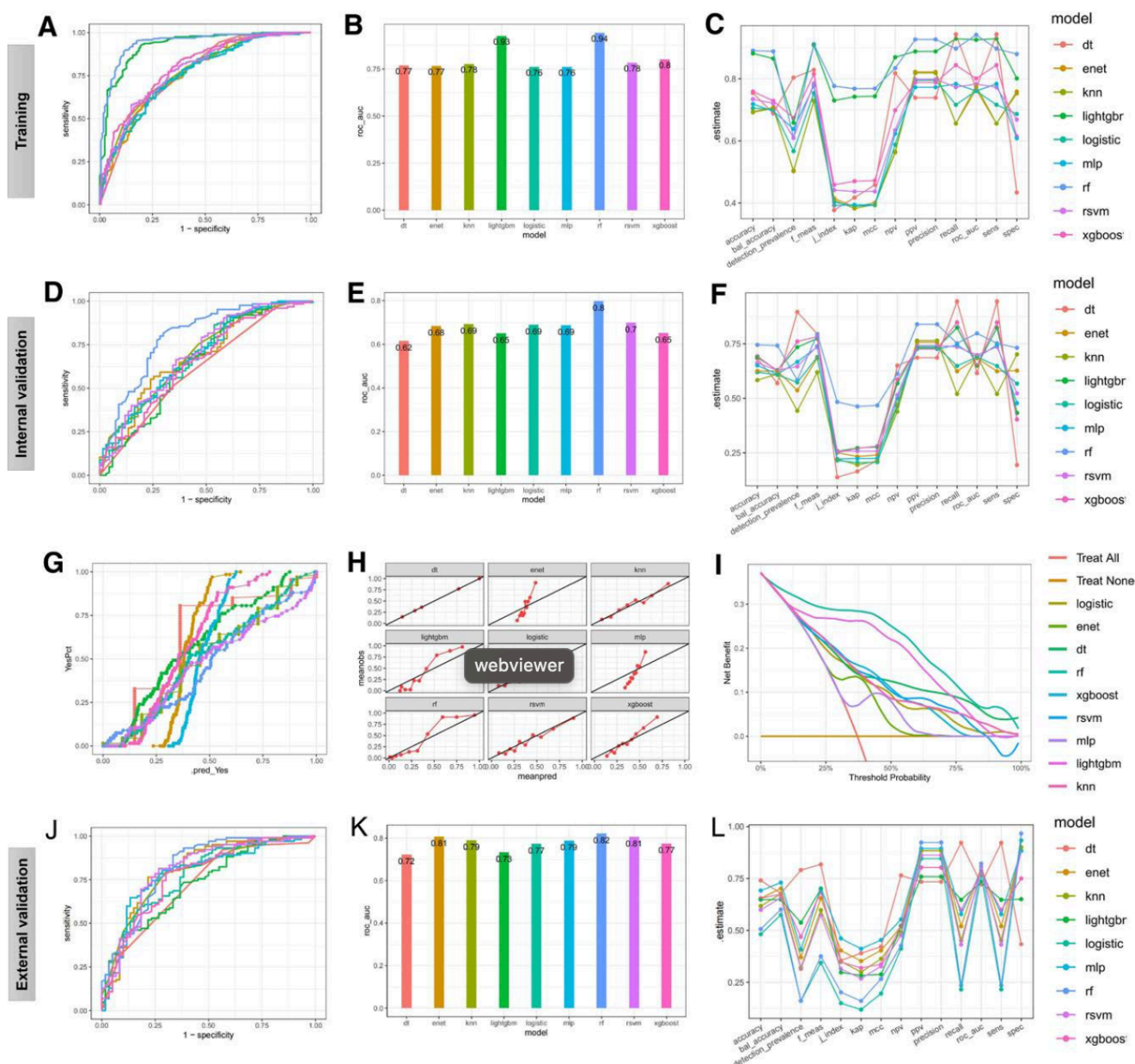


FIGURE 3. Performance comparison of 9 machine learning models on training, internal validation, and external validation cohorts. A, D, and J, ROC curves for each model in training, internal validation, and external validation cohorts, respectively. B, E, and K, AUC comparison of each model in training, internal validation, and external validation cohorts, respectively. C, F, and L, Diagnostic performance for each model in training, internal validation, and external validation cohorts, respectively. G, Cumulative calibration curves for each model. H, Segmented calibration curves for each model. I, Comparison of DCA results for each model. AUC, area under curve; DCA, decision curve analysis; DT, decision tree; Enet, elastic net; Kap, kappa coefficient; KNN, k-nearest neighbor; Lightgbm, light gradient boosting machine; Logistic, logistic regression; MCC, Matthews correlation coefficient; MLP, multilayer perceptron; NPV, positive predictive value; PPV, negative predictive value; RF, random forest; ROC, receiver operating characteristic; RSVM, radial support vector machine; Sens, sensitivity; Spec, specificity; Xgboost, extreme gradient boosting.

for evaluating model discrimination ability in training and internal validation cohorts are demonstrated in Figure 3C and F and Table S4 (SDC, <http://links.lww.com/TP/D224>). In the training cohort, the RF model exhibited an accuracy of 0.8906, a sensitivity of 0.8972, a specificity of 0.8795, an F1 score of 0.9117, and an AUC value of 0.9415 for PGD3 prediction (Table S4, SDC, <http://links.lww.com/TP/D224>). It also performed as expected in the internal validation cohort, with an accuracy of 0.7448, a sensitivity of 0.7520, a specificity of 0.7313, an F1 score of 0.7932, and an AUC value of 0.7975 (Table S4, SDC, <http://links.lww.com/TP/D224>). These comparative analyses indicated

that the RF model possessed the best performance, outperforming the other 8 models across various parameters. Additionally, cumulative calibration curves (Figure 3G) and segmented calibration curves (Figure 3H) revealed a robust alignment of RF model predictions with observations, and decision curve analysis curves underscored the RF model preeminent clinical value among the evaluated approaches (Figure 3I). Ultimately, we evaluated the generalization ability of the models, discovering that the RF model still demonstrated the largest AUC value among all models in the external validation cohort, with an accuracy of 0.7654, a sensitivity of 0.8235, a specificity of 0.6667, an F1 score of

0.8155, and an AUC value of 0.8214 (Figure 3J–L; Table S4, SDC, <http://links.lww.com/TP/D224>).

Risk Stratification for Postoperative Secondary Outcomes

Aside from accurately predicting primary outcomes such as PGD3, it is also crucial for prediction models to effectively stratify the risks associated with secondary outcomes. Our study indicated that the presence of PGD3 necessitated prolonged ECMO support, ventilatory assistance, and ICU management compared with non-PGD3 patients (Table 1). Given this, we further used 3 secondary outcomes, including postoperative ECMO support duration, MV duration, and ICU stay duration, to evaluate the stratification ability of each model. As shown in Table 2, except for DT and MLP, most models were capable of risk stratification. Notably, the RF model demonstrated the best ability to perform risk stratification, with statistically significant differences in ECMO time ($P < 0.001$), MV time ($P = 0.006$), and ICU stay time ($P < 0.001$; Table 2).

Model Interpretation

An analysis of the evolution of error rates in RF modeling for PGD3 revealed that the out-of-bag error rate steadily declined as more trees were added and ultimately stabilized around 100 trees (Figure 4A), indicating further increasing the number of trees would not reduce the error significantly but would increase computational cost without a corresponding gain in performance. Thus, 100 trees were selected as a reasonable trade-off between model performance and computational efficiency. Concurrently, the error rates for both positive and negative classes decreased, demonstrating enhanced classification performance on both types of samples (Figure 4A). Considering clinicians' requirement for explainable prediction models, the SHAP

algorithm was used to interpret the final model output by evaluating each variable contribution to PGD3 prediction. Utilization of the SHAP algorithm emphasized the predictive importance of the selected variables in the RF model to ensure a comprehensive understanding. Impact of each feature was depicted by distinctively colored dots, with red dots signifying larger risk values and blue dots representing smaller risk values (Figure 4B). Feature importance was further measured by accuracy decrease when permuted (mean decrease accuracy) and Gini impurity reduction in tree splits (mean decrease Gini). We found blood loss volume, HBP, and chronic steroid use ranked highly in both metrics, indicating their strong predictive power, whereas CIT and OT showcased higher Gini importance, suggesting their main contributions were through interactions (Figure 4C and D). Together, the dual metrics provided insights into the direct and interactive effects of each variable on the PGD3 prediction model. Aside from a global explanation that can elucidate overall functionality of the model, we visualized local explanation at the individual level. The comparisons between actual values and SHAP values for 9 feature variables are depicted in Figure 5. Notably, SHAP values >0 indicated a positive class prediction, implying an increased risk of PGD3. In general, more blood loss, CIT, and OT were suggestive of increased risk, whereas Lung Tx history, HBP, chronic steroid use, allogeneic transfusion (especially RBCs), and intratransplant ECMO support (especially venoarterial) were linked to increased SHAP levels.

DISCUSSION

We constructed and validated a clinically useful RF model to predict PGD3 incidence based on common clinical parameters using ML algorithms. This established prediction model may assist in posttransplant decision-making and care. Notably, RF, this predictive ML algorithm, displayed desirable discrimination, calibration,

TABLE 2.
Comparison of postoperative support based on risk stratification of different models

Model	Stratification	ECMO time, d	P (ECMO time)	MV time, d	P (MV time)	ICU time, d	P (ICU time)
DT	Low risk	1.75 ± 3.46	0.140	5.43 ± 13.43	0.303	8.38 ± 10.97	0.993
	High risk	2.19 ± 3.68		6.65 ± 15.35		8.38 ± 10.67	
Enet	Low risk	1.66 ± 3.40	0.004	4.52 ± 9.90	0.002	7.57 ± 10.02	0.044
	High risk	2.48 ± 3.79		8.22 ± 18.64		9.31 ± 11.50	
KNN	Low risk	1.53 ± 3.15	<0.001	4.49 ± 9.04	0.002	7.46 ± 9.73	0.030
	High risk	2.56 ± 3.96		8.04 ± 18.69		9.31 ± 11.66	
Lightgbm	Low risk	1.56 ± 2.91	<0.001	4.77 ± 9.42	0.004	7.52 ± 9.08	0.012
	High risk	2.99 ± 4.55		9.15 ± 21.40		10.06 ± 13.33	
Logistic	Low risk	1.43 ± 2.65	<0.001	4.76 ± 9.70	0.010	7.69 ± 9.89	0.012
	High risk	2.68 ± 4.31		7.79 ± 18.47		9.09 ± 11.57	
MLP	Low risk	1.44 ± 2.70	<0.001	5.39 ± 14.57	0.079	7.46 ± 10.04	0.012
	High risk	2.93 ± 4.49		7.48 ± 14.92		9.70 ± 11.62	
RF	Low risk	1.65 ± 3.04	<0.001	4.99 ± 11.19	0.006	7.27 ± 8.92	<0.001
	High risk	3.20 ± 4.72		9.89 ± 21.64		11.59 ± 14.43	
RSVM	Low risk	1.36 ± 2.45	<0.001	4.29 ± 9.28	0.002	7.40 ± 9.68	0.042
	High risk	2.55 ± 4.20		7.69 ± 17.60		9.10 ± 11.46	
Xgboost	Low risk	1.64 ± 3.03	<0.001	4.80 ± 9.66	0.005	7.66 ± 9.56	0.031
	High risk	2.85 ± 4.44		9.10 ± 21.24		9.79 ± 12.72	

DT, decision tree; ECMO, extracorporeal membrane oxygenation; ICU, intensive care unit; KNN, K-nearest neighbor; MLP, Multilayer Perceptron; MV, mechanical ventilation; RF, Random Forest; RSVM, Radial Support Vector Machine.

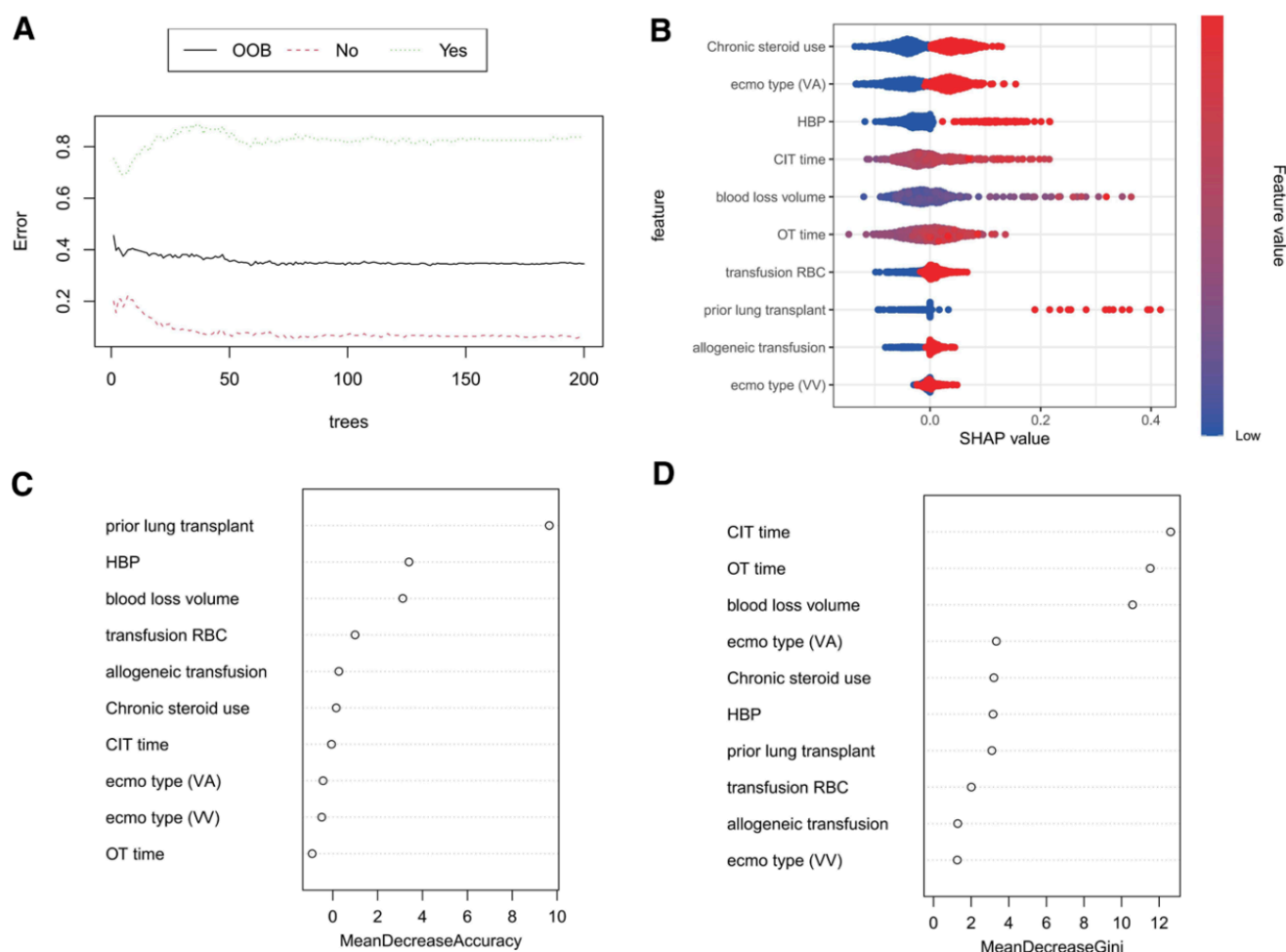


FIGURE 4. Construction and evaluation of the RF model for PGD3. A, The evolution of error rates (OOB, positive and negative) in RF modeling. B, Feature visualization of SHAP analysis. The x-axis represents the SHAP values, which quantify the contribution of each feature to the prediction of PGD3. C and D, Feature importance ranking indicated by mean decrease accuracy and mean decrease Gini. CIT, cold ischemia time; ECMO, extracorporeal membrane oxygenation; HBP, high blood pressure; OOB, out-of-bag; OT, operative time; PGD3, grade 3 primary graft dysfunction; RBC, red blood cell; RF, random forest; SHAP, SHapley Additive exPlanations; VA, venoarterial cannulation; VV, venovenous cannulation.

and net benefits in our cohorts. Meanwhile, some important issues, such as multicollinearity and overfitting, often overlooked in ML studies, were also addressed in our study.

The incidence and development of PGD3 have always been deemed as a poorly prognostic signal for patients after Lung Tx, in terms of whether short- or long-term outcomes,¹⁶ and early accurate diagnosis and timely effective interventions are crucial to improve survival prognosis and quality of life of patients. PGD is inherently a complicated, inflammatory syndrome, and its pathogenesis and risk factors are multifaceted.^{17,18} Clinically, patients with PGD3 will possibly develop with respiratory failure after Lung Tx. The occurrence of PGD might be attributed to a series of clinical factors.^{19–22} Our data showed that 233 patients (36.4%) ended up with PGD3, which is consistent with previously reported data.^{7,16} Moreover, several known risk factors were replicated by integrating univariate logistic regression and LASSO regression analyses in this study, including HBP, blood loss volume, CIT time, OT time, and RBC transfusion.^{8,13,21,23} In short, the PGD incidence rate and clinical high-risk parameters screened herein were similar to previous studies, supporting the clinical reliability and validity of our prediction model.

Since the incidence of PGD is determined by numerous factors, the predictive effect of a single clinical indicator is inherently inadequate, which has been underscored by an extensive body of investigations.^{7,24,25} Interestingly, several recent studies have revealed certain irregular biomarkers, such as blood lactate levels, cell-free DNA in ex vivo lung perfusate, and cell death proteins, as single biomarkers to effectively predict the occurrence of PGD3.^{10,26,27} However, for the latter 2 biomarkers, the prohibitive costs associated with sequencing and detection have impeded their transition from bench to bedside. Consequently, developing prediction models based on multiple parameters that are commonly used or easily collected in clinical practice may be the way out. Diamond et al¹¹ used regularized (lasso) logistic regression and established the PGD prediction model based on recipient, donor, and preoperative characteristics, and their PGD predictive algorithm and model were clinically useful with a user-friendly, publicly available interface for real-time risk assessment. In our study, we integrated the univariate logistic regression and LASSO regression analyses to conduct feature selection, which ultimately resulted in the screening of 9 selected variables out of >30 common clinical parameters associated with Lung Tx recipients or transplant operations. The accessibility of these commonly

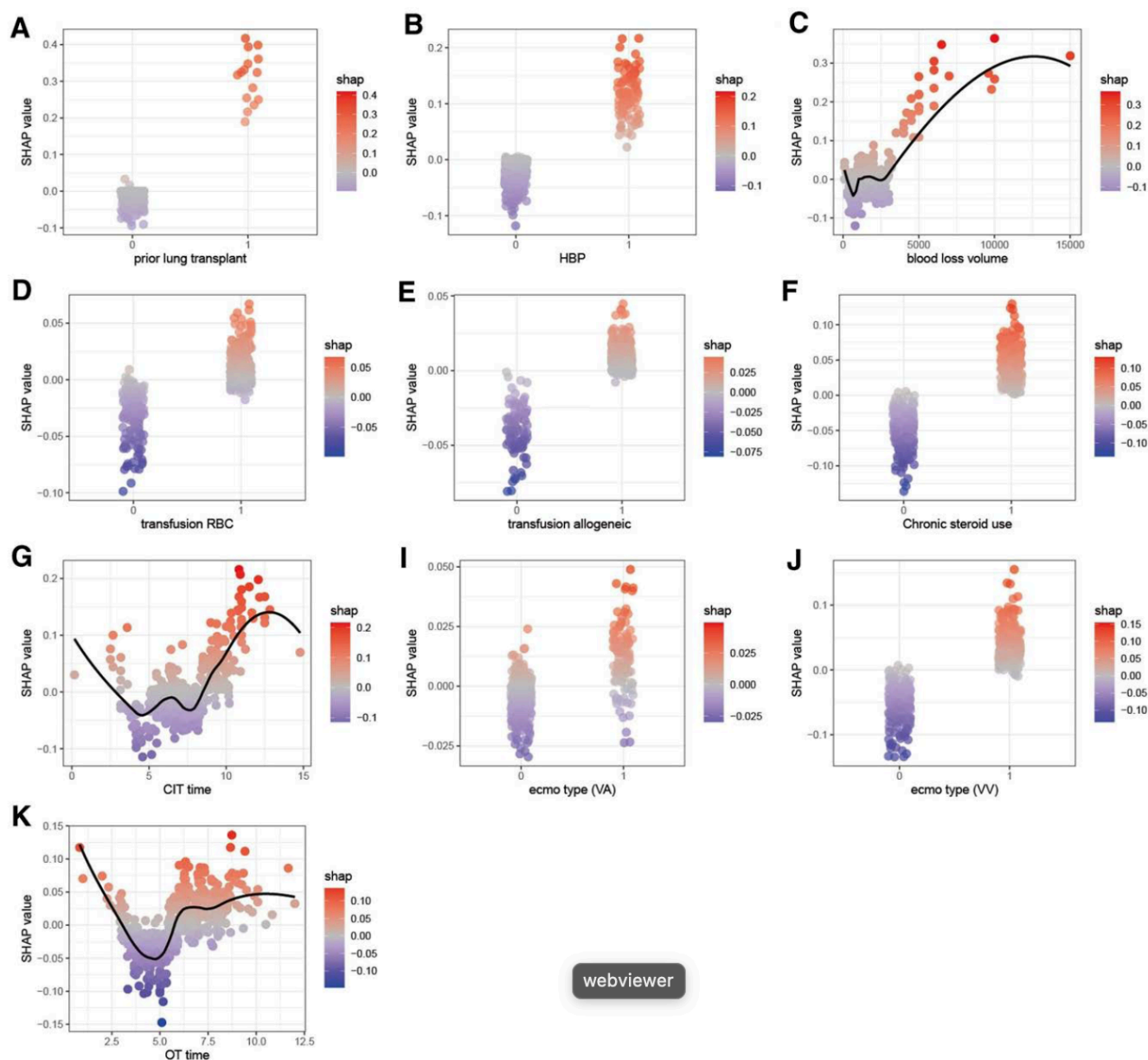


FIGURE 5. The SHAP dependence plot for each variable: prior lung transplant (A), HBP (B), blood loss (C), RBC transfusion (D), allogeneic transfusion (E), chronic steroid use (F), CIT (G), ECMO type (V-A) (H), ECMO type (V-V) (I), and OT (J). CIT, cold ischemia time; ECMO, extracorporeal membrane oxygenation; HBP, high blood pressure; OT, operative time; RBC, red blood cell; SHAP, SHapley Additive exPlanations; VA, venoarterial cannulation; VV, venovenous cannulation.

used parameters laid a foundation for the adaptability and applicability of our established prediction model.

Considering the constrained ability of traditional regression analysis methods in model construction, we used ML algorithms to develop prediction models based on selected variables. As artificial intelligence has rapidly evolved in recent years, ML has been fully applied to task-driven classification and outcome prediction, particularly in the field of surgery.²⁸ Particularly, in the field of organ transplantation, ML algorithms are often first considered to construct prediction models and evaluate patient prognosis based on high-dimensional data-driven model training and validation in a rigorous manner.²⁹ To date, only 2 studies have evaluated the utility of ML in predicting patient outcomes after Lung Tx.^{12,13} Michelson et al¹² used 4 prediction algorithms—logistic, KNN, Xgboost, and support vector classifier—to construct models to predict PGD3 based on

11 selected variables, and the KNN model had a relatively low AUC of 0.65, although showcasing superior calibration and performance than other models. In our study, a larger volume of ML algorithms, with the aforementioned 4 algorithms included, were adopted to establish precise prediction models based on 9 selected variables. Notably, the RF model surpassed the other 8 models, showcasing relatively higher sensitivity, specificity and AUC across training, internal validation, and external validation cohorts. Further assessments on calibration and clinical usefulness also indicated its promising applicability in PGD3 prediction. In addition, most ML models, except for DT and MLP, displayed potential in risk stratification for postoperative ECMO support duration, MV duration, and ICU stay duration. Similarly, the RF model still performed best, which also substantiated its latent applicability in clinical scenarios. Together, our study marched beyond

the identification of individual risk factors for PGD and applied ML algorithms to successfully construct an RF model for PGD3 prediction, which has desirable precision and calibration with promising clinical application.

Aside from prediction efficacy, interpretability of medical prediction models is equally important for the understanding, acceptance, and decision-making of clinicians.³⁰ We collected a range of clinical indicators associated with recipient characteristics and perioperative treatment to ensure the selected variables inputted as model parameters were relevant to patients' PGD3 outcomes. Meanwhile, model interpretability was further examined by the SHAP algorithm. In addition to measuring the predictive importance of selected variables, the SHAP algorithm can elucidate nonlinear relationships within risk indicators, providing global and local explanations.^{30,31} Globally, it could yield consistent feature attribution values to elucidate input-outcome relationships for PGD3, whereas locally, it provided a specific prediction for an individual patient. In our study, blood loss volume, HBP, and chronic steroid use were regarded to possess stronger predictive power because of their high feature importance rankings in both metrics, whereas CIT and OT demonstrated preeminent Gini importance, highlighting their contributions in terms of variable interaction. Altogether, the RF model is clinically explainable and will assist Lung Tx physicians in accurately predicting PGD3 incidence and performing risk stratification for patients after Lung Tx.

Our study introduces several innovative facets that distinguish it from previous endeavors. First, it focused on the Chinese population, which is a typical representative of Asian populations, as opposed to traditional studies on European and American populations. Second, few studies have used ML algorithms to study PGD prediction. We used 9 advanced ML algorithms instead of traditional univariate or multivariate logistic regression methods. Third, we developed an interpretable prediction model using the SHAP algorithm. The prominent model interpretability may help clinicians to better understand and make decisions, but this aspect has been largely unexplored in the realm of Lung Tx research. Nevertheless, there are several limitations inherent in our study. First, this study was designed as a single-center retrospective study with a limited sample size, thereby possibly resulting in a selection bias during case inclusion. The model discrimination may be very different in a much larger population. Second, the clinical characteristics collected were primarily related to the recipient and transplant procedure, and some potential clinical parameters that may affect the prognosis of Lung Tx patients were not fully considered in this study. Third, the established prediction models were tested only in a temporal-level external validation cohort in the absence of external validation from other centers, with their generalization ability not completely guaranteed. Last, the incorporated patients were all adults receiving Lung Tx. It remained undetermined whether the developed models can be applied clinically to predict the prognosis of pediatric patients after Lung Tx.

Collectively, the RF model showcased optimal performance in PGD3 prediction and risk stratification and has the potential to identify high-risk recipients and guide treatments for PGD3. We believe this established prediction model has important utility in the real world. It holds great promise in accurate and personalized prediction for patients with PGD3

after LTx and in posttransplant patient-level decision-making assistance for Lung Tx surgeons. Moving forward, future efforts will concentrate on conducting a large-sample multicenter study recruiting external and prospective validation cohorts in different regions and populations. Meanwhile, incorporating other clinical parameters (eg, computed tomography scan images, genetic data) into the prediction model or leveraging more advanced ML algorithms may predict PGD3 incidence for Lung Tx patients more accurately.

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